MTH2003 Feedback Sheet 2

These questions are designed for "formative assessment". They are to check your understanding. You should be able to do them on your own, after having done the examples sheet and the tutorial sheet. Your solutions to these questions should be submitted as a pdf via ELE by 12:00 on Friday of week 5 of the term. These questions will be marked by the tutor of your group tutorial, and feedback given to you.

1. Consider the following differential equation

$$y'' + 8y' + 25y = 800t\cos(5t), \quad y(0) = 4, \quad y'(0) = -16\,.$$

[i] Find the complementary solution.
[ii] Find a particular solution.
[iii] Write down the general solution.
[iv] Also find the solution satisfying the initial conditions.

2. Consider the following homogeneous equation

$$(1-t^2)\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2t\frac{\mathrm{d}y}{\mathrm{d}t} - 2y = 0 \quad t \neq \pm 1\,.$$

[i] Show that $y_1(t) = t$ is a solution of the equation.
[ii] Substitute $y(t) = v(t)y_1(t)$, where $v(t)$ is a new unknown function, into the differential equation. Obtain an equation for v' and hence find the general solution for v, the general solution for y, and from there identify the second linearly independent solution $y_2(t)$.
[iii] Substitute your expression for y_2 into the differential equation to check that it is a solution.
[iv] Verify, by calculating the Wronskian of $y_1(t)$ and $y_2(t)$, that solution $y_2(t)$ is linearly independent of $y_1(t)$.

3. Solve the following second-order linear nonhomogeneous equation using the Method of Variation of Parameters:

$$t^2y'' - 4ty' + 6y = 21t^{-4}, \quad t \neq 0.$$